

Passive Pre-Association Wi-Fi Access-Point Selection

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Abstract— A Wi-Fi client in overlapping coverage must select an access point (AP) before it can directly measure the service that each AP would provide. Signal strength describes link quality, but not contention or load, and may therefore favor a nearby but congested AP. This project investigates whether a client can make a better decision using only passive observations collected before association, without probe requests or assistance from the network. This report defines the experimental foundation of that investigation: a controlled ns-3 simulator that creates multi-AP deployments, records decoded frames and physical carrier-sense activity, and replays each fixed topology once per target AP, changing only the AP the candidate joins. Every replay of a topology therefore shares one identical pre-association observation while yielding its own measured throughput. The simulator separates physical-topology randomness from radio randomness and produces a structured choice-set dataset. Across 1500 choice sets, a gradient-boosted model over 35 passive features selects the best AP in 93.4% of cases, against 64.7% for strongest signal strength and 86.7% for a heuristic that also accounts for channel occupancy. Sequence models over the same observation do not improve on this, and the report sets out what a further iteration would need for the value of temporal structure to become testable.

I. PROBLEM FORMULATION

Throughput depends on channel contention, competing stations, transmission rates, retries, and the airtime consumed by neighboring basic service sets. Received signal strength reflects such effects poorly if at all. Those effects do, however, leave traces a client can hear without transmitting: the frames themselves, and the time the medium spends busy. This project asks whether such traces are enough: *can a client choose the AP that will provide the greatest throughput using only what its radio can passively observe before association?*

Existing load-aware methods [1], [2] make their load estimate after association, from AP-provided reports, or from a network controller. A client making its first choice has none of these: only beacons and some data frames, in which AP distance and offered traffic appear only through their effects.

Let $\mathcal{A} = \{a_1, \dots, a_N\}$ be the APs in one deployment and let O be the candidate station's passive observation before it joins any AP. If y_i is the throughput obtained after joining a_i , the desired decision is

$$a^* = \arg \max_{a_i \in \mathcal{A}} y_i, \quad (1)$$

using a decision function whose inputs are derived only from O .

We can tackle this problem either by creating a model that solely ranks the available choices to determine a^* , or by simply predicting the value of y_i for each a_i .

We attempt both methods to learn of their efficacy in this project as both are possible with a simulated environment. In a live network however, only one y_i is ever measurable: the station joins a single AP and observes that AP's throughput, while the values of y_j for every AP it did not join are counterfactual. The simulator recovers all N values by replaying the same scenario once per target AP, changing only the AP the candidate joins therefore granting every choice set in the generated data the true value of a^* .

One key assumption must be carefully verified in the simulated data: O must be identical across all N runs of a given scenario. If any part of the observation varied with the AP eventually joined, a decision function could reach the right answer through a simulation artifact rather than through the network.

II. SIMULATOR DESIGN AND ASSUMPTIONS

The simulator is built on ns-3.45 and its Wi-Fi models [8]. The choice set presented to the candidate station contains only APs discovered during its passive scan.

A. Simulation Environment

The deployment is static and single-floor with no walls, the environment TGax assumes for its indoor small-BSS scenario [5]. All stations are immobile and remain at their seeded positions. Radios are homogeneous IEEE 802.11n devices at 5 GHz on 20 MHz channels with Minstrel-HT rate control [8].

Every radio is an ns-3 `SpectrumWifiPhy` attached to one shared `MultiModelSpectrumChannel`, on which a log-distance loss (exponent 3.0, with 46.6777 dB of reference loss at 1 m) is followed by Nakagami fading.

`SpectrumWifiPhy` is chosen over the simpler `YansWifiPhy` as the Yans model simulates exactly zero cross channel interference by treating each channel as an isolated layer. If `YansWifiPhy` is used, the channel occupancy given by CCA measurements becomes a really powerful signal for models in a way that does not reflect reality. `SpectrumWifiPhy`, on the other hand, represents a transmission as a power spectral density shaped by the 802.11 transmit mask, so a neighbor one channel away deposits real power in a receiver's band and registers in its carrier sense.

B. Count, Topology and Channel occupancy of APs

Each scenario first draws 2, 3, 4, 6, or 8 APs. They sit on a triangular lattice 20 m apart: TGax recommends “~10–20 m inter AP distance” for its dense indoor scenarios [5]. Two APs form a pair, three an equilateral triangle, and the four, six, and eight AP cases fill two rows of two, three, and four with alternate rows offset by half a spacing, so the nearest-neighbor distance does not change as options are added.

Finally, each AP draws a channel uniformly at random from channels 36, 40, 44, and 48. Random assignment biases the data toward more in-channel contention and interference which could potentially make the problem harder than if we plan out the channels to be minimally interfering. However, TGax sets precedent for random assignment across four channels in its enterprise scenario, and across the non-overlapping set in its residential one [5], so this is the strategy adopted.

C. Count, Topology of background STAs

TODO: Rewrite this subsection. The placement description is verbose, and the rationale for mixing uniform and hotspot regimes is neither obvious nor established.

A base count of 3, 4, or 5 is drawn independently for each AP and the draws are summed. That sum is the deployment’s station total, not an allocation: how many stations an AP ends up serving follows from where they are placed and which AP they associate with, not from its own draw.

A quarter of scenarios spread the stations uniformly. The rest declare one or more APs as “hotspots” around which stations are clustered. Such mixing of uniform and deliberately clustered regimes follows common practice in WLAN studies [3], [4], and is of particular import here because under uniform load every AP is about equally busy and the problem we examine collapses heavily to signal strength.

We draw the number of hotspots at random with fewer hotspots being more likely and the count capped at $\lceil N_{AP}/3 \rceil$. The area around a hotspot AP, a 10 m disc around it, is assigned a probability density 4 times that of non-hotspot areas. Each station’s position is drawn from this distribution.

D. Association and Load of background STAs

Each station associates with the AP whose received power is highest. That power comes from the log-distance term alone, perturbed by an independent Gaussian of 5 dB standard deviation drawn once per station–AP pair. Without the perturbation stations form exact Voronoi partitions, which no real deployment does; 5 dB is the indoor shadowing deviation TGax specifies [5]. The perturbation is confined to this decision. The simulated channel carries no shadowing term of its own, so it stands in for the wall and obstruction variation a no-walls environment omits.

Every station then draws its own offered load, independently, from a log-normal distribution with median 2.25 Mbit/s and shape 0.9, capped at 25 Mbit/s, for a mean of 3.37 Mbit/s. Traffic volumes are log-normal rather than Gaussian [7], and the right skew is what puts most stations on a light load and a few on a heavy one without needing a mixture. The quartiles

span the mix an AP actually carries, from voice and browsing up through high-definition video.

E. Placement of candidate STA

The candidate station is placed uniformly at random over the same area as the background stations: the AP footprint plus a 10 m border.

F. Data Capture

Background traffic starts at $t = 0.5$ s. Each station’s arrivals are Poisson at its own data rate, and its first packet is delayed by a phase drawn uniformly over one mean interval. Without that phase, stations sharing an AP and a rate would transmit in lockstep for the whole run. The non-full-buffer models in the TGax set randomize phase for the same reason [5].

Passive recording runs from $t = 2$ s to $t = 8$ s, covering 6 s of loaded traffic. It starts 1.5 s after the traffic because Minstrel-HT needs about that long to converge, and until it does every scenario carries the same opening burst of retransmissions. This burst is a property of when the run started, not of the deployment being observed.

The candidate starts joining at $t = 8$ s, the moment recording ends.

After joining, the candidate offers 100 Mbit/s in 1250-byte packets until the run ends at $t = 20$ s. That rate keeps it backlogged, so its measured throughput reflects the service the network gives it rather than the limit of its own traffic generator.

G. Matched Counterfactual Replays

For each sampled topology and radio seed, the experiment runs once per target AP. AP and station positions, background traffic, the candidate’s position, and every pre-association device stay fixed; only the AP selected at $t = 8$ s changes. A `topologySeed` fixes the deployment: station placement, channel selection, and the load offered by each station.

An independent `rngSeed` controls everything that varies between one occasion and another — fading, contention, rate adaptation, and the arrival times of the background packets themselves. Three radio seeds are generated per topology.

A station’s demand is a property of the deployment; when its packets happen to leave is not, so only the data rate belongs to the first seed. Separating the two streams is what allows stochastic radio behavior to be repeated without the physical deployment shifting underneath it. This provides insight as to how much of the true answer is dependent purely on the topological configuration of stations and APs and how much depends on the random time evolution of the medium.

III. OBSERVATION AND DATASET CONSTRUCTION

A. Recording and Projection

A deployed client scans across all channels that host APs in a rotating fashion. The simulation records more than any deployed client can be expected to observe by recording activity on every occupied channel simultaneously. We then project this complete record down to what a single scan would see.

Specifically, the candidate STA carries one receive-only scanner PHY per occupied channel, plus a separate PHY

TABLE I

EVERY QUANTITY DRAWN FROM `TOPOLOGYSEED`, ONCE PER SCENARIO.

Quantity	Distribution
AP count	uniform on $\{2, 3, 4, 6, 8\}$
AP channel	uniform on $\{36, 40, 44, 48\}$, iid per AP
Station count	sum over APs of uniform on $\{3, 4, 5\}$
Spatial regime	uniform with prob. $1/4$, else hotspots
Hotspot count	$1 \dots \lceil N_{\text{AP}}/3 \rceil$, fewer more likely
Hotspot field	$4\times$ station density inside a 10 m disc
Station load	log-normal, median 2.25 Mbit/s, shape 0.9, cap 25
Association	strongest received power + $\mathcal{N}(0, 5^2)$ dB
Candidate position	uniform over the station area

reserved for association. The scanner PHYs never transmit and never associate; the association PHY is parked on unused channel 149 until the decision time. At $t = 8$ s the scanner PHYs move to the parking channel as well.

Two traces are recorded. The monitor-sniffer trace holds every frame a scanner decodes, with the start and end of the transmission carrying it, frequency, BSSID, transmitter, frame category, beacon and retry flags, length, RSSI, airtime, and PHY rate. Frames aggregated into one transmission share its start and end. The PHY-state trace gives the fraction of each 1 ms interval in which carrier sense marked the channel busy. This clear-channel-assessment (CCA) measurement registers energy a frame-only view misses: interference, and transmissions too distant to decode. Both thresholds are left at the ns-3 defaults: -62 dBm of received energy from any source, or -82 dBm from a signal whose preamble the PHY recognizes. Some driver stacks expose an equivalent channel-busy counter, though availability varies by platform.

The projection onto one passive radio is done as follows: One radio is assumed to rotate through the occupied channels in random order, dwelling 110 ms on each with 0.15 ms to retune, and repeats the cycle for as much of the window as it fits, so every channel is revisited many times before the decision. A frame is kept only if its whole transmission falls inside a dwell after retuning: a radio that misses the preamble decodes nothing. CCA intervals outside a dwell are discarded. A 110 ms dwell spans just over one 102.4 ms beacon interval, so each visit yields about one beacon from every AP on that channel. RSSI is then perturbed by uniform error within ± 3 dB and quantized to 1 dB. Both steps run after ns-3 from a recorded scan seed, so scan policy and measurement noise can be varied without repeating a simulation.

The recorded frame and channel-busy traces are verified to be byte-identical across all target-AP replays of a matched scenario to enforce the constant O required in Section I).

B. From Trace to Training Data

In order to train models of particular architectures, we must design appropriate representations of the raw captured data. The first representation we use is simply the raw trace itself: every decoded frame in order, carrying its arrival time, channel, BSSID and transmitter, class, flags, length, signal, airtime, and PHY rate. It is a time series, and we train a transformer on it

directly.

The raw observation is a long sequence, and what distinguishes one AP from another does not reside at frame granularity. Both problems that arise here are resolved by binning the time axis and aggregating quantities; As this is possible only for features that can be averaged over a set of frames, we define a list of hand crafted features. These are enumerated in Appendix A.

Aggregation is done per AP before the data is bucketed. Frames are attributed to an AP by BSSID. Time is bucketed by channel visit: one bin per dwell, so that no bin spans a retune and aggregates traffic from two channels into a figure representing neither. A 5.5 s window against a 110 ms dwell gives roughly 50 bins. Each bin carries two sets of quantities: one aggregated over that AP's own frames, the other over all traffic the radio heard during the visit. The second is shared by every AP in the choice set, since one radio hears one channel at a time; a flag records whether the channel being visited was that AP's own. A transformer attends over bins exactly as it does over frames.

Widening the bucket to the whole window removes time altogether and collapses everything into a single row of tabular data per AP (Table II). AP-specific quantities are aggregated over that AP's frames alone; channel-specific quantities over the channel that AP occupies. Gradient boosting, random forests, and ridge regression are trained on this representation.

TABLE II
WHAT EACH MODEL RECEIVES FOR ONE CHOICE SET OF N OPTIONS.

Form	Per option	Consumed by
Frames	the decoded trace, one token per frame	Transformer
Binned	~ 50 intervals $\times$ 29 measurements	Transformer
Vector	35 measurements	GBM, RF, ridge

Squashing the time axis in this way carries no comparison between options, since it aggregates over one AP alone: the row records how busy an AP is, not whether it is busier than its neighbors. A set of relative features is therefore engineered and appended, restoring for the tabular models the comparison that per-AP aggregation removed. A sequence or set model needs no such repair, because its set stage attends across options directly. The cost of leaving the repair out is visible in Table III: the same binned measurements averaged over the window, with no relative features, support 0.803 against 0.934 for the engineered vector, though the two differ in more than the comparison.

The five radio-seed realizations of a topology are near-duplicates, so train/test splits are cut by topology: letting two of them fall on opposite sides would inflate every result reported below.

IV. MODELS, PROTOCOL, AND RESULTS

The dataset holds 6449 AP options across 1500 choice sets drawn from 300 topologies, a median of 4 options per decision. The best option in a set yields 42.6 Mbit/s on average against 22.0 for one chosen at random, and the best and worst differ by 35.5 Mbit/s.

Every decision function below reads the same observation and differs only in how much of it it is handed. The comparison is between representations as much as architectures.

A. Baselines

Three passive baselines are evaluated alongside random selection:

- *Strongest RSSI* ranks options by mean beacon level.
- *Least busy channel* ranks them by the clear-channel-assessment busy fraction of the channel each occupies, so options sharing a channel are indistinguishable to it; tied options are scored at their mean throughput.
- *RSSI minus busy* maps RSSI from -100 to -60 dBm onto the unit interval and subtracts the busy fraction from it.

B. Learned Models

Each model reads one representation, and the groups in Table III say which. Ridge regression, random forests, gradient boosting, a pointwise MLP and a set ranker read the single-row vector. A separate set of tabular models reads the binned sequence instead. One transformer reads the binned sequence and another reads the frame trace; a third pair reads the vector and the binned sequence together, which is what measures the trajectory against the averages.

1) *Tabular Models*: On the vector a tabular model reads one row of 35 numbers per option. The binned sequence is not tabular, so it is reduced three ways before such a model can read it: averaged over time, summarised by six statistics per measurement (mean, spread, extremes, last value and least-squares slope), or handed over whole with bin order carried by column position. Table III names these three *averaged over time*, *bin statistics* and *all bins*. Gradient boosting and a pointwise MLP are fitted to each of the three, ridge to the flattened one only, every fit a separate model on a separate input. Both neural models are single hidden-layer networks; the set ranker scores an option against the rest of its choice set, the pointwise MLP scores each row alone.

Every model on the vector is selected per split by validation regret: ridge over regularization strength, gradient boosting over learning rate (0.03 to 0.1), leaf size (4 to 16) and 200 or 400 iterations, the random forest over leaf size and feature fraction at 400 trees, and the neural models over weight decay and score temperature. Both tree models also choose whether to fit a log-transformed target. On the sequence encodings gradient boosting is tuned the same way; the MLPs there use one fixed setting.

2) *Transformers*: Both transformers use a model dimension of 48, four attention heads, and dropout 0.1, and hold roughly 10^5 parameters. The binned model encodes each option's sequence with two self-attention layers over bins, reads out a learned summary token, then compares options with two further layers over the option set, masked so padding cannot contribute. The frame-level model cannot attend over the raw sequence directly, since a trace runs to thousands of frames; four learned queries per option cross-attend into it instead, which is linear rather than quadratic in frame count. Both are

trained with AdamW at 2×10^{-3} , weight decay 10^{-3} , gradient clipping at unit norm, and early stopping on validation regret.

C. Protocol

The learned models divide between two objectives. Regression predicts $\log(1 + \text{throughput})$ in Mbit/s, so its outputs carry a scale. Ranking learns a distribution over the options in a choice set and optimizes the decision directly, at the cost of scores that carry no scale. Only the transformers are run under both.

Each result averages five 60/20/20 partitions of the topologies, stratified by AP count and redrawn per seed, so test sets overlap and the spread is draw-to-draw variation rather than a cross-validation error. Every model sees the same partitions; the frame-level model was run on the first three. Neural models keep one initialization throughout, so their spread excludes initialization variance. Model selection uses regret against the oracle choice on the validation split; the reported metric is top-1 accuracy with a 0.5 Mbit/s tie tolerance.

TABLE III
TOP-1 ACCURACY, MEAN AND STANDARD DEVIATION OVER FIVE PARTITIONS, EXCEPT †, OVER THREE. GROUPS DESCEND THE AGGREGATION LADDER OF SECTION III. HEURISTICS ARE SCORED ON THE FEATURE-VECTOR PIPELINE.

Model	Obj.	Top-1
<i>Heuristics</i>		
Random choice	—	0.282 ± 0.002
Least busy channel	—	0.442 ± 0.072
Strongest RSSI	—	0.647 ± 0.042
RSSI minus busy	—	0.867 ± 0.038
<i>Frame trace</i>		
Transformer†	reg.	0.888 ± 0.039
Transformer†	rank	0.897 ± 0.028
<i>Binned sequence</i>		
Ridge, all bins	reg.	0.768 ± 0.030
MLP, bin statistics	rank	0.821 ± 0.035
GBM, bin statistics	reg.	0.842 ± 0.029
GBM, all bins	reg.	0.845 ± 0.020
MLP, all bins	rank	0.847 ± 0.032
Transformer	reg.	0.873 ± 0.026
Transformer	rank	0.873 ± 0.037
<i>Binned sequence, averaged over time</i>		
MLP	rank	0.767 ± 0.039
GBM	reg.	0.803 ± 0.059
<i>Vector: 35 hand-built features</i>		
Ridge	reg.	0.881 ± 0.045
MLP, pointwise	rank	0.897 ± 0.024
Set ranker	rank	0.909 ± 0.030
Random forest	reg.	0.925 ± 0.019
Gradient boosting	reg.	0.934 ± 0.020
<i>Vector and binned sequence</i>		
Transformer	reg.	0.919 ± 0.021
Transformer	rank	0.931 ± 0.030

D. Results

Every learned model beats every heuristic, so passive evidence past signal strength does carry something a client can use. Table III then reads down the ladder.

Aggregation is not what limits these models. The frame trace, aggregated not at all, supports 0.897; the binned sequence supports 0.873 under a transformer and 0.847 under the best tabular encoding; averaging those same bins to one row per option gives 0.803. Collapsing time costs 0.042 for gradient boosting and 0.080 for the pointwise MLP, the one comparison here where a single variable moves with the model held fixed.

The engineered vector reverses the ordering. It is also one row per option, yet gradient boosting reads 0.934 from it, ahead of random forests at 0.925 and of every model given the sequence. Paired by partition, gradient boosting beats the random forest on all five, by 0.000 to 0.020, so the ordering at the top holds even though the two standard deviations overlap. What separates 0.803 from 0.934 is not how much of the observation survives but which quantities were computed from it.

Giving a transformer the vector and the sequence together settles the remaining question. It reaches 0.931, where gradient boosting already sits with no sequence at all, so the trajectory adds nothing measurable to the averages.

V. INTERPRETATION AND NEXT STEPS

The results establish that passive evidence beyond signal strength is useful for AP selection. They do not establish that the order of that evidence matters, since a transformer given the vector and the sequence together lands where gradient boosting on the vector alone already sits.

The simulator gives temporal modeling little to exploit. After background traffic begins, the ratio of within-series to between-series standard deviation is 0.09 for frame rate, 0.09 for channel-busy fraction, and 0.17 for the option's own activity. Most measured variation lies between APs rather than within an AP over the window. An AP looks much the same at one second as at five. This is evidence that averaging loses little rather than proof that order is irrelevant, since a stationary-looking sequence can still carry autocorrelation, drift, or short-lived events.

The intraclass correlation of 0.988 is not evidence for either representation. It shows that throughput is highly reproducible once topology and target AP are fixed across radio seeds, not whether that throughput is best inferred from a time average, an ordered sequence, or something else.

One reason the sequence alone struggles is sparsity. Under a rotating radio roughly three-quarters of an option's bins carry no frames from it, because the radio was listening elsewhere. The vector is computed over the whole window and does not have that problem, which is a difference in coverage rather than in what either representation says about time.

The frame-level experiment points the same way. On the three splits both were evaluated on under the ranking objective, the frame transformer reaches 0.897 against the binned transformer's 0.883, a gap smaller than the spread either shows across splits. Keeping individual frames buys nothing clear, though only under traffic as steady as this.

The simulator supports the premise of the project: a client can infer useful information about AP quality from passive

pre-association observation, and learned combinations of that evidence beat signal strength alone by a wide margin. What it cannot show is whether order would matter in a network that moves, since this one barely does. The suggestions below are aimed at that.

A. Suggested Extensions

Two changes bear on the question this work leaves open. The first is to make the medium non-stationary on a timescale shorter than the observation window, and visibly so in a passive trace: load that arrives and departs, stations that join and leave, interference that appears part-way through. The second is to shorten the label. A twelve-second mean of an already-stationary process is predicted best by the window mean, whereas a label measured over the second or two after the decision gives a trend something to extrapolate toward. Together they would make the value of temporal order identifiable rather than assumed.

The observation itself could be made more representative. The 802.11 BSS Load element, which advertises a station count and channel utilisation and which a client reads from a single beacon, is absent here, and it is the strongest baseline a real passive client has. The window could also be defined by the pre-association period rather than by complete silence, admitting the probe requests most clients actually send.

Two aspects of the scenario generator would repay attention. Channels are assigned by AP index over a fixed layout, so co-channel separation has no variance and reuse is tied to deployment size. The effect is a synthetic shortcut rather than leakage in the usual sense, since the number of discovered options is legitimately observable: option count determines AP count in 91.8% of choice sets, and AP count fixes the channel plan exactly, so a model can read the reuse regime off the size of its input rather than off radio evidence. The guard of Section I checks only that the observation is independent of which AP is joined, so it does not catch this; sampling the channel assignment would. Separately, the 30 m spacing pairs an obstructed-indoor path-loss exponent with an open-field geometry.

B. Ranking Versus Regression

The two objectives are not equally testable outside a simulator, and the difference favors the one this work treats as secondary.

Ranking asks which AP is best. Establishing ground truth for that answer requires joining each option under identical conditions, which is exactly what the matched replay of Section II provides and what no field measurement can. A ranking model's output is therefore not easily verifiable per decision in the field. Regression on the other hand can be tested reliably for accurate predictions of throughput.

This work reports both objectives because the simulator supports both. Work aimed at deployment should treat throughput prediction as the primary target, and derive the choice from it, rather than optimizing the choice directly and inheriting a claim that cannot be checked.

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APPENDIX A HAND-BUILT FEATURES

The 35 columns available to the tabular models, derived from frames the candidate decoded before transmitting or associating and, for the CCA column, from its scanner’s carrier-sense state. Every rate and fraction is normalized by time actually spent listening to the relevant channel, not by the observation window.

TABLE IV

THE FEATURE COLUMNS. AP-LEVEL AND CHANNEL-LEVEL FEATURES DESCRIBE THE OPTION; RELATIVE FEATURES DESCRIBE IT AGAINST ITS RIVALS, AND EXIST TO RESTORE COMPARISONS THAT PER-AP FILTERING REMOVED.

Column	Meaning
AP-level	
ap_rssi_{mean, std, max, min, last}	Beacon RSSI in dBm: mean, spread, extremes, and most recent
ap_beacons	Beacons decoded
ap_beacon_gap_{mean, std}	Interval between decoded beacons, mean and spread; set by scan revisit, not beacon period
ap_frames{, _per_s}	Frames attributed to this BSS, total and per second listening
ap_bytes_per_s	Bytes per second attributed to this BSS
ap_airtime_frac	Fraction of listening time this BSS occupied
ap_n_clients	Distinct transmitters other than the AP
ap_data_frames	Data frames from this BSS
ap_mean_data_rate	Mean PHY rate of its data frames
ap_retry_frac	Fraction of its frames marked retry
Channel-level	
chan_busy_frac	Channel occupancy from decoded frame airtime
chan_cca_busy_frac	Channel occupancy from PHY carrier sense; counts energy never decoded
chan_frames_per_s	All frames per second on this channel
chan_bytes_per_s	All bytes per second
chan_{data, mgmt, ctrl}_frac	Share of frames that are data, management, control
chan_retry_frac	Share marked retry
chan_mean_data_rate	Mean PHY rate of data frames
chan_n_{bssids, tas}	Distinct BSSIDs and transmitters heard
chan_mean_rssi	Mean RSSI over all frames
Relative	
rel_n_options	Options in this choice set
rel_rssi_rank	Options strictly stronger than this one
rel_is_strongest	Whether no option is stronger
rel_rssi_margin_best_other	RSSI minus the best alternative’s
rel_rssi_minus_mean	RSSI minus the option-set mean
rel_airtime_share	Share of observed airtime; denominator spans every BSS heard, so it does not sum to one
rel_clients_share	Share of observed transmitters